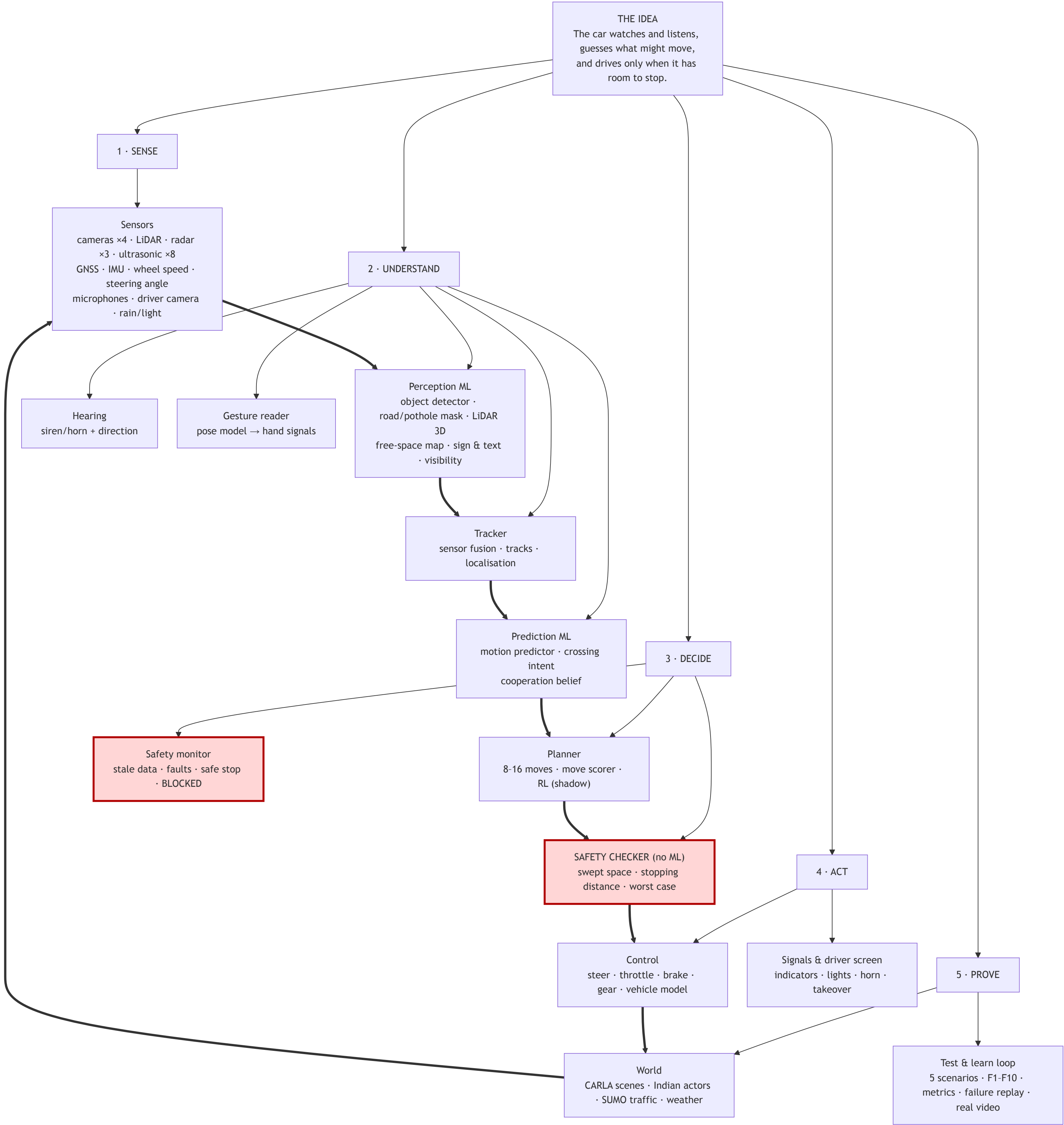


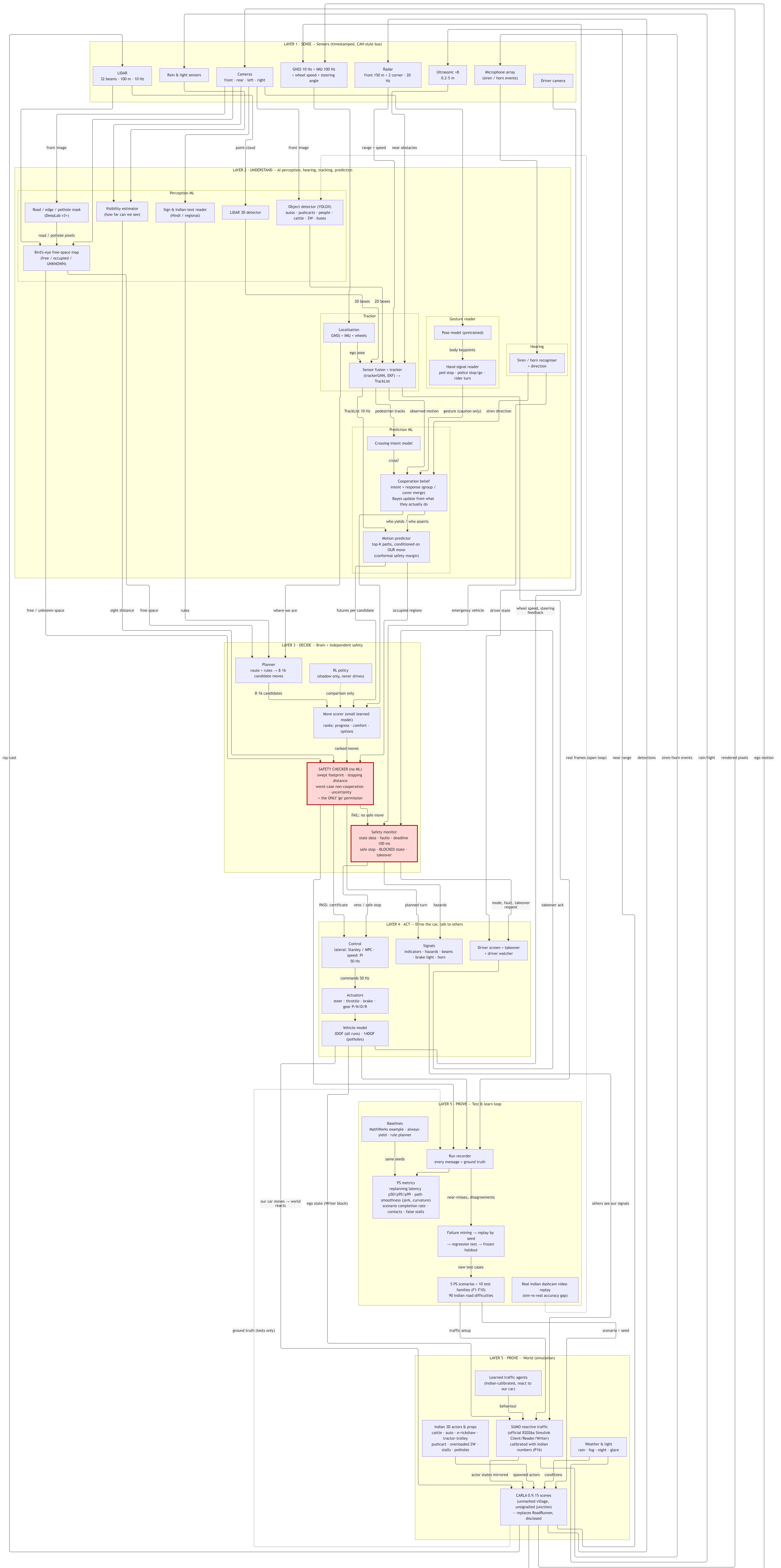
D1 · Full system — the hierarchy (1 idea → 5 layers → 13 parts)

Read top-down. Thick arrows = the main loop the car runs 10 times a second. Page 2 = every part and every connection.



D1 · Full system — end to end

1 idea → 5 layers (SENSE · UNDERSTAND · DECIDE · ACT · PROVE) → 13 parts → components. Red = safety (only the checker can say 'go'). Dashed = test / comparison paths.



D1 · Full system — the hierarchy (1 idea → 5 layers → 13 parts) — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

```
LEVEL 0: THE IDEA — "The car watches and listens, guesses what might move, and drives only when it has room to stop."
LEVEL 1 (5 layers): 1 SENSE · 2 UNDERSTAND · 3 DECIDE · 4 ACT · 5 PROVE
LEVEL 2 (13 parts):
  SENSE      → Sensors
  UNDERSTAND → Perception ML · Hearing · Gesture reader · Tracker · Prediction ML
  DECIDE     → Planner · Safety checker (no ML) · Safety monitor
  ACT        → Control · Signals & driver screen
  PROVE      → World · Test & learn loop
LEVEL 3: components listed inside each part box (see page 2 for the full wiring).
THICK LOOP (the main data path): Sensors → Perception ML → Tracker → Prediction ML → Planner → Safety checker → Control → World → back to Sensors.
```

Mermaid source (paste into mermaid.live to re-render)

```
flowchart TB
  IDEA["THE IDEA<br>The car watches and listens, guesses what might move,<br>and drives only when it has room to stop."]
  IDEA --> L1["1 · SENSE"] & L2["2 · UNDERSTAND"] & L3["3 · DECIDE"] & L4["4 · ACT"] & L5["5 · PROVE"]
  L1 --> p1["Sensors<br>cameras x4 · LiDAR · radar x3 · ultrasonic x8<br>GNSS · IMU · wheel speed · steering angle<br>microphones · driver camera · rain/light"]
  L2 --> p2["Perception ML<br>object detector · road/pothole mask · LiDAR 3D<br>free-space map · sign & text · visibility"]
  L2 --> p3["Hearing<br>siren/horn + direction"]
  L2 --> p4["Gesture reader<br>pose model → hand signals"]
  L2 --> p5["Tracker<br>sensor fusion · tracks · localisation"]
  L2 --> p6["Prediction ML<br>motion predictor · crossing intent<br>cooperation belief"]
  L3 --> p7["Planner<br>8–16 moves · move scorer · RL (shadow)"]
  L3 --> p8["SAFETY CHECKER (no ML)<br>swept space · stopping distance · worst case"]
  L3 --> p9["Safety monitor<br>stale data · faults · safe stop · BLOCKED"]
  L4 --> p10["Control<br>steer · throttle · brake · gear · vehicle model"]
  L4 --> p11["Signals & driver screen<br>indicators · lights · horn · takeover"]
  L5 --> p12["World<br>CARLA scenes · Indian actors · SUMO traffic · weather"]
  L5 --> p13["Test & learn loop<br>5 scenarios · F1–F10 · metrics · failure replay · real video"]
  p1 ==> p2
  p2 ==> p5
  p5 ==> p6
  p6 ==> p7
  p7 ==> p8
  p8 ==> p10
  p10 ==> p12
  p12 ==> p1
  classDef safety fill:#ffd6d6,stroke:#b00000,stroke-width:3px
  class p8,p9 safety
```

D1 · Full system — end to end — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

BOXES (grouped exactly as drawn):

- LAYER 5 · PROVE – World (simulation)
 - [W_carla] CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed
 - [W_assets] Indian 3D actors & props cattle · auto · e-rickshaw · tractor-trolley pushcart · overloaded 2W · stalls · potholes
 - [W_sumo] SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16)
 - [W_agents] Learned traffic agents (Indian-calibrated, react to our car)
 - [W_weather] Weather & light rain · fog · night · glare
- LAYER 1 · SENSE – Sensors (timestamped, CAN-style bus)
 - [S_cam] Cameras front · rear · left · right
 - [S_lidar] LiDAR 32 beams · 100 m · 10 Hz
 - [S_radar] Radar front 150 m + 2 corner · 20 Hz
 - [S_us] Ultrasonic x8 0.2–5 m
 - [S_loc] GNSS 10 Hz + IMU 100 Hz + wheel speed + steering angle
 - [S_mic] Microphone array (siren / horn events)
 - [S_dcam] Driver camera
 - [S_env] Rain & light sensors
- LAYER 2 · UNDERSTAND – AI perception, hearing, tracking, prediction
 - Perception ML
 - [M_det] Object detector (YOLOX) autos · pushcarts · people · cattle · 2W · buses
 - [M_road] Road / edge / pothole mask (DeepLab v3+)
 - [M_lidar3d] LiDAR 3D detector
 - [M_bev] Bird's-eye free-space map (free / occupied / UNKNOWN)
 - [M_sign] Sign & Indian-text reader (Hindi / regional)
 - [M_vis] Visibility estimator (how far can we see)
 - Hearing
 - [M_audio] Siren / horn recogniser + direction
 - Gesture reader
 - [M_pose] Pose model (pretrained)
 - [M_gest] Hand-signal reader ped stop · police stop/go · rider turn
 - Tracker
 - [T_fuse] Sensor fusion + tracker (trackerGNN, EKF) → TrackList
 - [T_loc] Localisation GNSS + IMU + wheels
 - Prediction ML
 - [M_pred] Motion predictor top-K paths, conditioned on OUR move (conformal safety margin)
 - [M_intent] Crossing-intent model
 - [B_belief] Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do
- LAYER 3 · DECIDE – Brain + independent safety
 - [D_plan] Planner route + rules → 8–16 candidate moves
 - [D_score] Move scorer (small learned model) ranks: progress · comfort · options
 - [D_rl] RL policy (shadow only, never drives)
 - [D_check] SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission
 - [D_mon] Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover
- LAYER 4 · ACT – Drive the car, talk to others
 - [A_ctrl] Control lateral: Stanley / MPC · speed: PI 50 Hz
 - [A_act] Actuators steer · throttle · brake · gear P/N/D/R
 - [A_veh] Vehicle model 3DOF (all runs) · 14DOF (potholes)
 - [A_sig] Signals indicators · hazards · beams · brake light · horn
 - [A_hmi] Driver screen + takeover + driver watcher
- LAYER 5 · PROVE – Test & learn loop
 - [E_log] Run recorder every message + ground truth
 - [E_metrics] PS metrics replanning latency p50/p95/p99 · path smoothness (jerk, curvature) scenario completion rate · contacts · false stalls
 - [E_scen] 5 PS scenarios + 10 test families (F1–F10) 90 Indian road difficulties
 - [E_mine] Failure mining → replay by seed → regression test → frozen holdout
 - [E_real] Real Indian dashcam video replay (sim-vs-real accuracy gap)
 - [E_base] Baselines MathWorks example · always-yield · rule planner

CONNECTIONS (73 arrows, from → to : what flows):

- CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed → Cameras front · rear · left · right : rendered pixels
- CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed → LiDAR 32 beams · 100 m · 10 Hz : ray-cast
- CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed → Radar front 150 m + 2 corner · 20 Hz : detections
- CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed → Ultrasonic x8 0.2–5 m : near range
- CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed →

GNSS 10 Hz + IMU 100 Hz + wheel speed + steering angle : ego motion
 SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16) → Microphone array (siren / horn events) : siren/horn events
 Weather & light rain · fog · night · glare → Rain & light sensors : rain/light
 Indian 3D actors & props cattle · auto · e-rickshaw · tractor-trolley pushcart · overloaded 2W · stalls · potholes → CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed : spawned actors
 SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16) → CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed : actor states mirrored
 Learned traffic agents (Indian-calibrated, react to our car) → SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16) : behaviour
 Weather & light rain · fog · night · glare → CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed : conditions
 Cameras front · rear · left · right → Object detector (YOLOX) autos · pushcarts · people · cattle · 2W · buses : front image
 Cameras front · rear · left · right → Road / edge / pothole mask (DeepLab v3+) : front image
 Cameras front · rear · left · right → Sign & Indian-text reader (Hindi / regional)
 Cameras front · rear · left · right → Pose model (pretrained)
 Cameras front · rear · left · right → Visibility estimator (how far can we see)
 Rain & light sensors → Visibility estimator (how far can we see)
 LiDAR 32 beams · 100 m · 10 Hz → LiDAR 3D detector : point cloud
 LiDAR 32 beams · 100 m · 10 Hz → Bird's-eye free-space map (free / occupied / UNKNOWN)
 Cameras front · rear · left · right → Bird's-eye free-space map (free / occupied / UNKNOWN)
 Radar front 150 m + 2 corner · 20 Hz → Sensor fusion + tracker (trackerGNN, EKF) → TrackList : range + speed
 Ultrasonic ×8 0.2–5 m → Sensor fusion + tracker (trackerGNN, EKF) → TrackList : near obstacles
 GNSS 10 Hz + IMU 100 Hz + wheel speed + steering angle → Localisation GNSS + IMU + wheels
 Microphone array (siren / horn events) → Siren / horn recogniser + direction
 Driver camera → Driver screen + takeover + driver watcher : driver state
 Object detector (YOLOX) autos · pushcarts · people · cattle · 2W · buses → Sensor fusion + tracker (trackerGNN, EKF) → TrackList : 2D boxes
 LiDAR 3D detector → Sensor fusion + tracker (trackerGNN, EKF) → TrackList : 3D boxes
 Road / edge / pothole mask (DeepLab v3+) → Bird's-eye free-space map (free / occupied / UNKNOWN) : road / pothole pixels
 Pose model (pretrained) → Hand-signal reader ped stop · police stop/go · rider turn : body keypoints
 Localisation GNSS + IMU + wheels → Sensor fusion + tracker (trackerGNN, EKF) → TrackList : ego pose
 Sensor fusion + tracker (trackerGNN, EKF) → TrackList → Motion predictor top-K paths, conditioned on OUR move (conformal safety margin) : TrackList 10 Hz
 Sensor fusion + tracker (trackerGNN, EKF) → TrackList → Crossing-intent model : pedestrian tracks
 Sensor fusion + tracker (trackerGNN, EKF) → TrackList → Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do : observed motion
 Crossing-intent model → Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do : cross?
 Hand-signal reader ped stop · police stop/go · rider turn → Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do : gesture (caution only)
 Siren / horn recogniser + direction → Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do : siren direction
 Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do → Motion predictor top-K paths, conditioned on OUR move (conformal safety margin) : who yields / who asserts
 Bird's-eye free-space map (free / occupied / UNKNOWN) → Planner route + rules → 8–16 candidate moves : free space
 Sign & Indian-text reader (Hindi / regional) → Planner route + rules → 8–16 candidate moves : rules
 Localisation GNSS + IMU + wheels → Planner route + rules → 8–16 candidate moves : where we are
 Planner route + rules → 8–16 candidate moves → Move scorer (small learned model) ranks: progress · comfort · options : 8–16 candidates
 Motion predictor top-K paths, conditioned on OUR move (conformal safety margin) → Move scorer (small learned model) ranks: progress · comfort · options : futures per candidate
 Cooperation belief intent × response (group / cover merge) Bayes update from what they actually do → Move scorer (small learned model) ranks: progress · comfort · options
 RL policy (shadow only, never drives) → Move scorer (small learned model) ranks: progress · comfort · options : comparison only
 Move scorer (small learned model) ranks: progress · comfort · options → SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission : ranked moves
 Motion predictor top-K paths, conditioned on OUR move (conformal safety margin) → SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission : occupied regions
 Bird's-eye free-space map (free / occupied / UNKNOWN) → SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission : free / unknown space
 Visibility estimator (how far can we see) → SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission : sight distance
 SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission → Control lateral: Stanley / MPC · speed: PI 50 Hz : PASS: certificate

SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission → Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover : FAIL: no safe move

Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover → Control lateral: Stanley / MPC · speed: PI 50 Hz : veto / safe stop

Siren / horn recogniser + direction → Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover : emergency vehicle

Driver screen + takeover + driver watcher → Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover : takeover ack

Control lateral: Stanley / MPC · speed: PI 50 Hz → Actuators steer · throttle · brake · gear P/N/D/R : commands 50 Hz

Actuators steer · throttle · brake · gear P/N/D/R → Vehicle model 3DOF (all runs) · 14DOF (potholes)

Vehicle model 3DOF (all runs) · 14DOF (potholes) → GNSS 10 Hz + IMU 100 Hz + wheel speed + steering angle : wheel speed, steering feedback

Vehicle model 3DOF (all runs) · 14DOF (potholes) → CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed : our car moves → world reacts

Vehicle model 3DOF (all runs) · 14DOF (potholes) → SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16) : ego state (Writer block)

SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission → Signals indicators · hazards · beams · brake light · horn : planned turn

Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover → Signals indicators · hazards · beams · brake light · horn : hazards

Safety monitor stale data · faults · deadline 100 ms safe stop · BLOCKED state · takeover → Driver screen + takeover + driver watcher : mode, fault, takeover request

Signals indicators · hazards · beams · brake light · horn → SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16) : others see our signals

Sensor fusion + tracker (trackerGNN, EKF) → TrackList → Run recorder every message + ground truth

SAFETY CHECKER (no ML) swept footprint · stopping distance worst-case non-cooperation · uncertainty = the ONLY 'go' permission → Run recorder every message + ground truth

Vehicle model 3DOF (all runs) · 14DOF (potholes) → Run recorder every message + ground truth

CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed → Run recorder every message + ground truth : ground truth (tests only)

Run recorder every message + ground truth → PS metrics replanning latency p50/p95/p99 · path smoothness (jerk, curvature) scenario completion rate · contacts · false stalls

Run recorder every message + ground truth → Failure mining → replay by seed → regression test → frozen holdout : near-misses, disagreements

Failure mining → replay by seed → regression test → frozen holdout → 5 PS scenarios + 10 test families (F1–F10) 90 Indian road difficulties : new test cases

5 PS scenarios + 10 test families (F1–F10) 90 Indian road difficulties → CARLA 0.9.15 scenes (unmarked village, unsignalled junction) – replaces RoadRunner, disclosed : scenario + seed

5 PS scenarios + 10 test families (F1–F10) 90 Indian road difficulties → SUMO reactive traffic (official R2026a Simulink Client/Reader/Writer) calibrated with Indian numbers (P16) : traffic setup

Real Indian dashcam video replay (sim-vs-real accuracy gap) → Object detector (YOLOX) autos · pushcarts · people · cattle · 2W · buses : real frames (open loop)

Baselines MathWorks example · always-yield · rule planner → PS metrics replanning latency p50/p95/p99 · path smoothness (jerk, curvature) scenario completion rate · contacts · false stalls : same seeds

Mermaid source (paste into mermaid.live to re-render)

```

graph TD
    subgraph L5W ["LAYER 5 · PROVE – World (simulation)"]
        W_carla["CARLA 0.9.15 scenes<br/>(unmarked village, unsignalled junction)<br/>– replaces RoadRunner, disclosed"]
        W_assets["Indian 3D actors & props<br/>cattle · auto · e-rickshaw · tractor-trolley<br/>pushcart · overloaded 2W · stalls · potholes"]
        W_sumo["SUMO reactive traffic<br/>(official R2026a Simulink Client/Reader/Writer)<br/>calibrated with Indian numbers (P16)"]
        W_agents["Learned traffic agents<br/>(Indian-calibrated, react to our car)"]
        W_weather["Weather & light<br/>rain · fog · night · glare"]
    end

    subgraph L1 ["LAYER 1 · SENSE – Sensors (timestamped, CAN-style bus)"]
        S_cam["Cameras<br/>front · rear · left · right"]
        S_lidar["LiDAR<br/>32 beams · 100 m · 10 Hz"]
        S_radar["Radar<br/>front 150 m + 2 corner · 20 Hz"]
        S_us["Ultrasonic x8<br/>0.2–5 m"]
        S_loc["GNSS 10 Hz + IMU 100 Hz<br/>+ wheel speed + steering angle"]
        S_mic["Microphone array<br/>(siren / horn events)"]
        S_dcam["Driver camera"]
        S_env["Rain & light sensors"]
    end

    subgraph L2 ["LAYER 2 · UNDERSTAND – AI perception, hearing, tracking, prediction"]
        subgraph P ["Perception ML"]
            M_det["Object detector (YOLOX)<br/>autos · pushcarts · people · cattle · 2W · buses"]
            M_road["Road / edge / pothole mask<br/>(DeepLab v3+)"]
            M_lidar3d["LiDAR 3D detector"]
        end
    end

```

```

    M_bev["Bird's-eye free-space map<br/>(free / occupied / UNKNOWN)"]
    M_sign["Sign & Indian-text reader<br/>(Hindi / regional)"]
    M_vis["Visibility estimator<br/>(how far can we see)"]
end
subgraph H["Hearing"]
    M_audio["Siren / horn recogniser<br/>+ direction"]
end
subgraph G["Gesture reader"]
    M_pose["Pose model (pretrained)"]
    M_gest["Hand-signal reader<br/>ped stop · police stop/go · rider turn"]
end
subgraph T["Tracker"]
    T_fuse["Sensor fusion + tracker<br/>(trackerGNN, EKF) → TrackList"]
    T_loc["Localisation<br/>GNSS + IMU + wheels"]
end
subgraph PR["Prediction ML"]
    M_pred["Motion predictor<br/>top-K paths, conditioned on OUR move<br/>(conformal safety margin)"]
    M_intent["Crossing-intent model"]
    B_belief["Cooperation belief<br/>intent × response (group / cover merge)<br/>Bayes update from
what they actually do"]
end
subgraph L3["LAYER 3 · DECIDE – Brain + independent safety"]
    D_plan["Planner<br/>route + rules → 8–16 candidate moves"]
    D_score["Move scorer (small learned model)<br/>ranks: progress · comfort · options"]
    D_rl["RL policy<br/>(shadow only, never drives)"]
    D_check["SAFETY CHECKER (no ML)<br/>swept footprint · stopping distance<br/>worst-case non-
cooperation · uncertainty<br/>= the ONLY 'go' permission"]
    D_mon["Safety monitor<br/>stale data · faults · deadline 100 ms<br/>safe stop · BLOCKED state ·
takeover"]
end
subgraph L4["LAYER 4 · ACT – Drive the car, talk to others"]
    A_ctrl["Control<br/>lateral: Stanley / MPC · speed: PI<br/>50 Hz"]
    A_act["Actuators<br/>steer · throttle · brake · gear P/N/D/R"]
    A_veh["Vehicle model<br/>3DOF (all runs) · 14DOF (potholes)"]
    A_sig["Signals<br/>indicators · hazards · beams · brake light · horn"]
    A_hmi["Driver screen + takeover<br/>+ driver watcher"]
end
subgraph L5["LAYER 5 · PROVE – Test & learn loop"]
    E_log["Run recorder<br/>every message + ground truth"]
    E_metrics["PS metrics<br/>replanning latency p50/p95/p99 · path smoothness (jerk, curvature)
<br/>scenario completion rate · contacts · false stalls"]
    E_scen["5 PS scenarios + 10 test families (F1–F10)<br/>90 Indian road difficulties"]
    E_mine["Failure mining → replay by seed<br/>→ regression test → frozen holdout"]
    E_real["Real Indian dashcam video replay<br/>(sim-vs-real accuracy gap)"]
    E_base["Baselines<br/>MathWorks example · always-yield · rule planner"]
end
W_carla -->|"rendered pixels"| S_cam
W_carla -->|"ray-cast"| S_lidar
W_carla -->|"detections"| S_radar
W_carla -->|"near range"| S_us
W_carla -->|"ego motion"| S_loc
W_sumo -->|"siren/horn events"| S_mic
W_weather -->|"rain/light"| S_env
W_assets -->|"spawned actors"| W_carla
W_sumo -->|"actor states mirrored"| W_carla
W_agents -->|"behaviour"| W_sumo
W_weather -->|"conditions"| W_carla
S_cam -->|"front image"| M_det
S_cam -->|"front image"| M_road
S_cam --> M_sign
S_cam --> M_pose
S_cam --> M_vis
S_env --> M_vis
S_lidar -->|"point cloud"| M_lidar3d
S_lidar --> M_bev
S_cam --> M_bev
S_radar -->|"range + speed"| T_fuse
S_us -->|"near obstacles"| T_fuse
S_loc --> T_loc
S_mic --> M_audio
S_dcaml -->|"driver state"| A_hmi
M_det -->|"2D boxes"| T_fuse
M_lidar3d -->|"3D boxes"| T_fuse
M_road -->|"road / pothole pixels"| M_bev
M_pose -->|"body keypoints"| M_gest
T_loc -->|"ego pose"| T_fuse
T_fuse -->|"TrackList 10 Hz"| M_pred
T_fuse -->|"pedestrian tracks"| M_intent

```

```

T_fuse -->|"observed motion"| B_belief
M_intent -->|"cross?"| B_belief
M_gest -->|"gesture (caution only)"| B_belief
M_audio -->|"siren direction"| B_belief
B_belief -->|"who yields / who asserts"| M_pred
M_bev -->|"free space"| D_plan
M_sign -->|"rules"| D_plan
T_loc -->|"where we are"| D_plan
D_plan -->|"8-16 candidates"| D_score
M_pred -->|"futures per candidate"| D_score
B_belief --> D_score
D_rl -->|"comparison only"| D_score
D_score -->|"ranked moves"| D_check
M_pred -->|"occupied regions"| D_check
M_bev -->|"free / unknown space"| D_check
M_vis -->|"sight distance"| D_check
D_check -->|"PASS: certificate"| A_ctrl
D_check -->|"FAIL: no safe move"| D_mon
D_mon -->|"veto / safe stop"| A_ctrl
M_audio -->|"emergency vehicle"| D_mon
A_hmi -->|"takeover ack"| D_mon
A_ctrl -->|"commands 50 Hz"| A_act
A_act --> A_veh
A_veh -->|"wheel speed, steering feedback"| S_loc
A_veh -->|"our car moves → world reacts"| W_carla
A_veh -->|"ego state (Writer block)"| W_sumo
D_check -->|"planned turn"| A_sig
D_mon -->|"hazards"| A_sig
D_mon -->|"mode, fault, takeover request"| A_hmi
A_sig -->|"others see our signals"| W_sumo
T_fuse --> E_log
D_check --> E_log
A_veh --> E_log
W_carla -->|"ground truth (tests only)"| E_log
E_log --> E_metrics
E_log -->|"near-misses, disagreements"| E_mine
E_mine -->|"new test cases"| E_scen
E_scen -->|"scenario + seed"| W_carla
E_scen -->|"traffic setup"| W_sumo
E_real -->|"real frames (open loop)"| M_det
E_base -->|"same seeds"| E_metrics
classDef safety fill:#ffd6d6,stroke:#b00000,stroke-width:3px
class D_check,D_mon safety

```